

# 80 GHz C+L Band Germanium Photodetector via Microheater

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**Abstract** A germanium photodetector integrated with microheater is demonstrated for operation in C+L band. The detector has a measured dark current of 30 nA and responsivity of 0.76 A/W at 1625 nm. The bandwidth is 80 GHz supporting 96 Gbps NRZ signalling at 1600 nm.

## Introduction

The fiber optical communications system is evolving towards extended C+L band for significantly increased transmission capacity [1,2]. This trend requires the transceiver to operate in a much wider wavelength range. As a proven photonic integration platform [3], silicon photonics has been witnessed to achieve great success for manufacturing transceivers in O band and C band. As silicon photonics evolve towards extended L band, most of the existing C band components can be easily redesigned to shift their central wavelengths to L band [4].

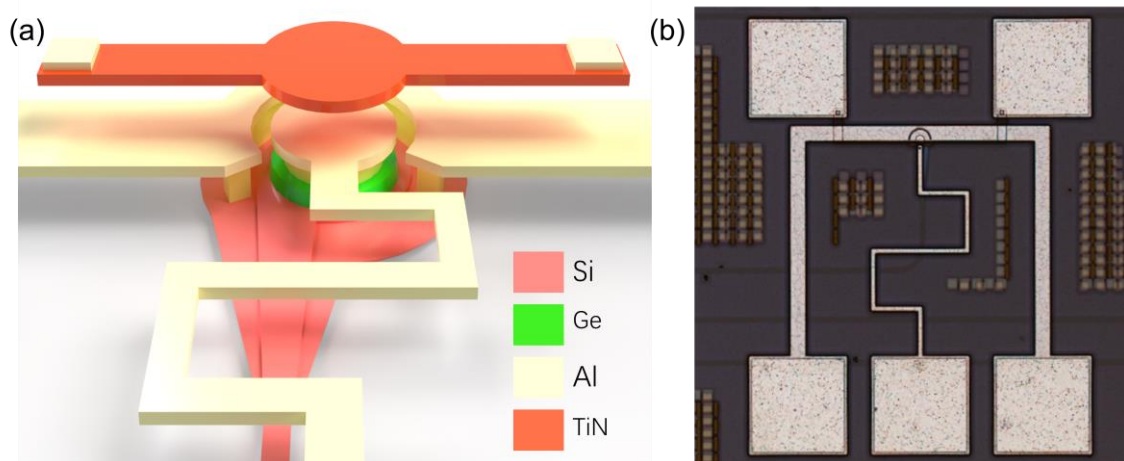
However, germanium is primarily used for photodetectors in silicon photonic transceiver, in which intrinsic absorption is almost forbidden for most of the wavelengths in L band. Heterogeneous integration of narrower bandgap materials is the most straightforward way to increase the intrinsic absorption at L band [5]. But introducing additional materials is not compatible with CMOS fabrication process. Cavity enhanced light matter interaction is also able to compensate the weak absorption [6]. But the resonant nature fundamentally limits the optical bandwidth. Similarly,

avalanche photodetectors can be used to improve the responsivity at longer wavelengths while the bandwidth is normally constrained [7]. While strain engineering is another possible way to shrink the bandgap of germanium, the strain is difficult to precisely control [8].

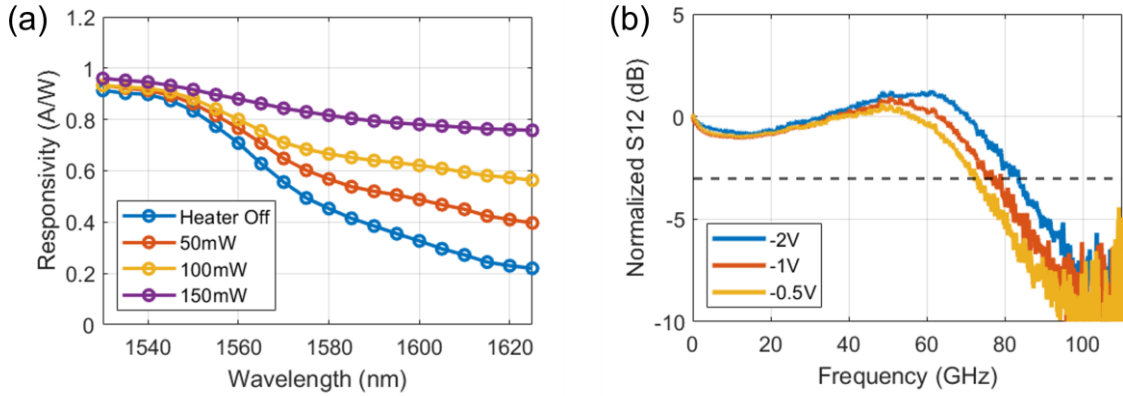
Here, we demonstrate a CMOS-compatible approach to enhance L-band absorption using on-chip thermal tuning without introducing additional materials or complex fabrication steps. The on-chip local heating effect can be used to shrink the bandgap of germanium and increases the absorption of photons beyond wavelength of 1600 nm. The device can be fabricated via commercial silicon photonic foundry without dedicate modification to the standard process. It offers a promising approach to extend the operating wavelength range of high-speed germanium photodetector, which might be useful for the emerging C+L band fiber optical communications.

## Device Structure

The device structure is illustrated in Fig. 1(a). Light is coupled from a silicon waveguide into a doughnut-shaped germanium (Ge) absorption region through evanescent coupling. The ring-



**Fig. 1:** (a) Three-dimensional schematic of the device. (b) Optical micrograph of the fabricated device.



**Fig. 2:** (a) Responsivity vs. wavelength at 1 mW optical input. (b) Normalized  $S_{12}$  responses at different bias voltages.

shaped geometry forms a compact light-trapping structure that effectively extends the optical absorption path while maintaining a reduced device footprint. Compared with conventional straight waveguide photodetectors, this configuration enables enhanced absorption efficiency without increasing the junction capacitance, thereby improving the bandwidth–responsivity trade-off. The device adopts a vertical p–i–n configuration to ensure efficient carrier collection under reverse bias.

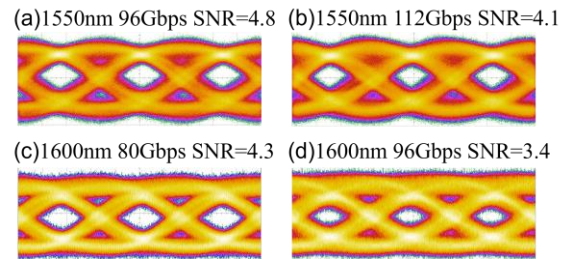
A titanium nitride (TiN) micro-heater is integrated above the Ge layer to enable thermal tuning of the absorption spectrum. By increasing the device temperature, the Ge bandgap is reduced, leading to an extension of the cutoff wavelength and enhanced absorption in the L-band. The heater is electrically isolated from the photodetector and connected through metal vias, allowing independent thermal control without affecting the RF signal path. This approach provides a CMOS-compatible and fabrication-friendly solution to overcome the intrinsic responsivity limitation of Ge in the L-band.

In addition, a meander inductor is incorporated into the signal electrode to introduce inductive peaking. This electrical design compensates for the RC-limited bandwidth by partially offsetting the capacitive roll-off at high frequencies. As a result, the frequency response is extended without significantly increasing the device complexity. The co-optimization of optical confinement, thermal tuning, and electrical bandwidth enhancement enables high-speed operation with improved L-band performance.

### Experimental Results

The device is fabricated via commercial silicon photonic foundry without dedicated process. The measured dark current is well below 30 nA at  $-2$  V when 150 mW electrical power applied to the heater. The measured responsivity and small-signal performance of the photodetector are shown in Fig. 2. The responsivity exceeds 0.9

A/W in the C-band and drops to  $\sim 0.2$  A/W at 1625 nm without heating. By applying 150 mW heating power, the responsivity at 1625 nm can be boosted to 0.76 A/W. This improvement clearly verifies the effectiveness of thermal tuning in extending the absorption edge of germanium beyond its intrinsic cutoff wavelength. The normalized  $S_{12}$  response shows a clear inductive peaking effect, with a 3 dB bandwidth of 82 GHz at  $-2$  V. Under thermal tuning, the bandwidth slightly decreases to 80 GHz, which is attributed to increased carrier scattering and longer transit time at elevated temperature.



**Fig. 3:** Eye diagrams at (a) 96 Gbps and (b) 112 Gbps under 1550 nm illumination; eye diagrams at (c) 80 Gbps and (d) 96 Gbps under 1600 nm illumination.

High-speed experimental results are presented in Fig. 3. At 1550 nm, clear and open eye diagrams are obtained at 112 Gbps with 2 dBm optical input. At 1600 nm, 96 Gbps NRZ signals are detected with an input power of  $-5$  dBm under 150 mW heating. The optical power incident to the detector at 1600 nm is much lower due to the limited gain of the erbium-doped fiber amplifier and increased modulator loss in the L-band. These results confirm the capability of the proposed photodetector for high-speed operation in C+L-band. A brief performance comparison summarized in Table 1 shows that this work achieves excellent overall performance in terms of dark current, responsivity and bandwidth.

**Tab. 1:** Performance comparison of representative Ge-on-Si photodetectors in C+L-band

|           | Dark Current<br>(nA) | Responsivity<br>(A/W) | Bandwidth<br>(GHz) | Data Rate<br>(Gbps) |
|-----------|----------------------|-----------------------|--------------------|---------------------|
| [6]       | 7.9                  | -                     | 25.7               | 40 NRZ @ 1600 nm    |
| [7]       | 4700                 | 1.16 @ 1600 nm        | 47.1               | 70 NRZ @ 1600 nm    |
| [8]       | 80                   | 0.83 @ 1625 nm        | 52.5               | -                   |
| [9]       | < 10                 | 1.1 @ 1620 nm         | > 50               | 200 PAM-4 @ 1610 nm |
| [10]      | 500                  | 0.71 @ 1620 nm        | > 110              | -                   |
| This Work | < 30                 | 0.76 @ 1625 nm        | 80                 | 96 NRZ @ 1600 nm    |

## Conclusion

In summary, we demonstrate a novel high-speed germanium photodetector for C+L band. The structure integrates the microheater for bandgap engineering and the metal inductor for inductive peaking. The device achieves 80 GHz bandwidth, 0.76 A/W responsivity at 1625 nm, and dark current below 30 nA. High-speed operation is validated by 96 Gbps NRZ reception at 1600 nm.

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